A-A-2114
March 17, 1983

COMMERCIAL ITEM DESCRIPTION

CASTER, RIGID (INDUSTRIAL DUTY)

The General Services Administration has authorized the use of this commercial item description in preference to type I, group B, style 4A of Federal Specification FF-C-88.

This description covers industrial quality rigid casters suitable for intermittent operation on nonpowered material transport equipment.

Salient characteristics

Casters furnished under this commercial item description shall have flat mounting plates, shall be of the size and height specified, and shall conform to the dimensions and load rating defined by figure 1 and table I.

Size B - 3-1/2 inches diameter wheel
Size C - 4 inches diameter wheel

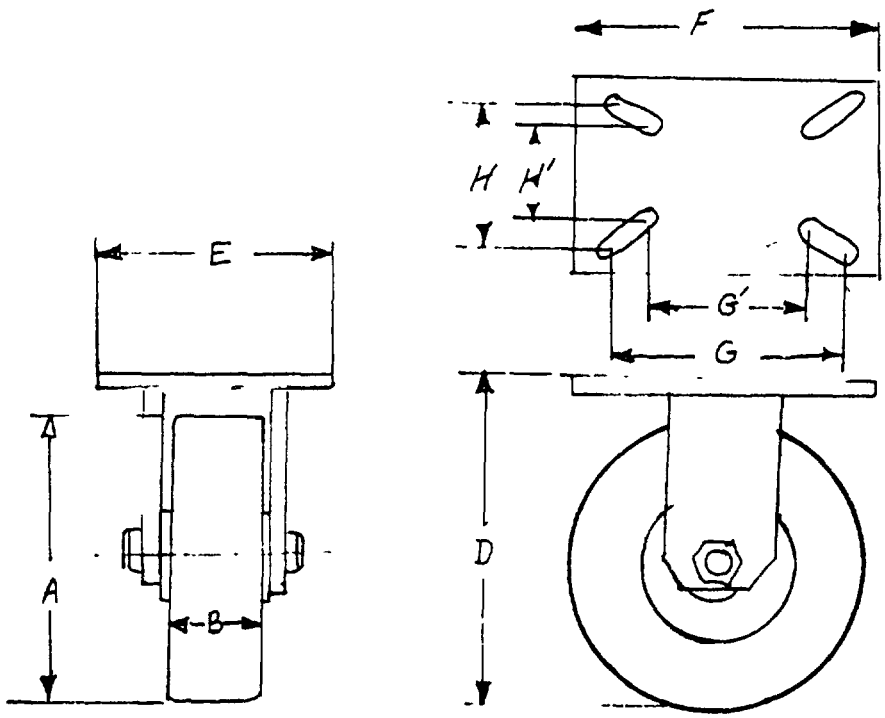


FIGURE I

TABLE I

Size	Load	B (min)	D ± 3/32	E ± 1/8	F ± 1/8	G ± 1/32	H ± 1/32
B	250	1-1/4	4-5/8	3-3/4	4-1/2	3 to 3-7/16*	2-3/8 to 3*
			4-19/32	2-1/2	5	4	1-3/16
			4-3/4	2-1/2	4-1/2	2-7/8 to 3*	1-3/4 to 2-3/8*
			4-1/2	2-1/2	3-5/8	2-7/8	1-3/4
C	400	1-1/2	5-1/4	3-1/2	6-1/2	5	1-3/4
		2	5-5/8	4	4-1/2	3 to 3-5/8*	2-5/8 to 3*

*Slotted

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Wheels. Casters shall have wheels which are of high quality hard rubber that is nonmarking, high impact, and oil resistant. The minimu n hardness shall be 80 ± 5 points on the D scale, Shore durometer in accordance with ASTM D 2240.

Wheel bearings. Casters shall be provided with roller bearings, ball bearings, or plain bearings as specified. When plain bearings are specified, they shall be formed of powdered metal with a high bronze content and they shall be fixed to the full width of the wheel hub. When roller bearings are specified, the outer raceway or sleeve shall be hardened steel.

Axle. The assembly shall be a 3/8 inch minimum diameter hex head steel bolt and nut. The axle shall be cold formed and shall be zinc or cadmium plated. During operation, the axle shall not rotate, loosen or bind the wheel.

Spanner bushing. The spanner bushing shall be of hardened steel and shall be ground or polished to form a smooth inner race for the bearing, minimum outside diameter 9/16 inch, minimum wall thickness 0 075 inch.

Thread guards. When thread guards are specified, the minimum wall thickness shall be 1/32 inch.

Horn assembly. The minimum wall thickness of the steel horn shall be 0 095 inch The horn shall engage the axle and spanner bushing to prevent rotation.

Rigid caster. The mounting plate and horn shall be an integral nonwelded unit or a welded cold forged assembly.

Lubrication. When specified the axle shall be prelubricated with water repellent grease having a -23° C to 71° C (-10° F to 160° F) operating range.

Lubrication fittings. When specified, the axle or wheel hub shall have pressure lubrication fittings.

Static load test. The caster shall withstand four times its rated capacity for 1-minute without damage.

Floor marking test. The caster, loaded to four times its rated capacity shall be cycled through six wheel revolutions in each direction. The smooth hardwood or concrete test surface shall show no evidence of treadmarks

Workmanship. The caster shall be free from imperfections which may impair durability or serviceability.

The issue of ASTM D 2240 in effect on the date of the solicitation shall be used to determine compliance with these requirements.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Preservation, packaging, packing, labeling, and marking The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes: The following options are available and should be specified in the contract or ordering documents.

(a) Size and height (dimension D of Figure 1) required.
(b) Type of wheel bearing required.
(c) If steel thread guards are required.
(d) If axle is to be prelubricated.
(e) If lubrication fittings are required.

This description does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this commercial item description, or those which are commercially available, but is intended to cover the types normally procured to meet Federal requirements

ASTM Standard D 2240 is available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 10103.

MILITARY INTERESTS:

Coordinating Activity

Army - AR

Custodians

Army - AR
Navy - YD
Air Force - 99

Review Activities

DSC - IS
Army - ME
Navy - MC

User Activities

Navy - YD, MC

CIVIL AGENCY COORDINATING ACTIVITY:

NASA - JFK
VA - VOC

PREPARING ACTIVITY:

GSA - FSS

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